

Geometry is the hidden decision behind every kernel method.

This book develops kernel methods from their mathematical foundations through approximation, learning theory, randomized computation, structured data, uncertainty, scientific machine learning, and modern operator methods. Proofs, algorithms, numerical checks, and geometric visualizations are designed to be read as one connected argument.

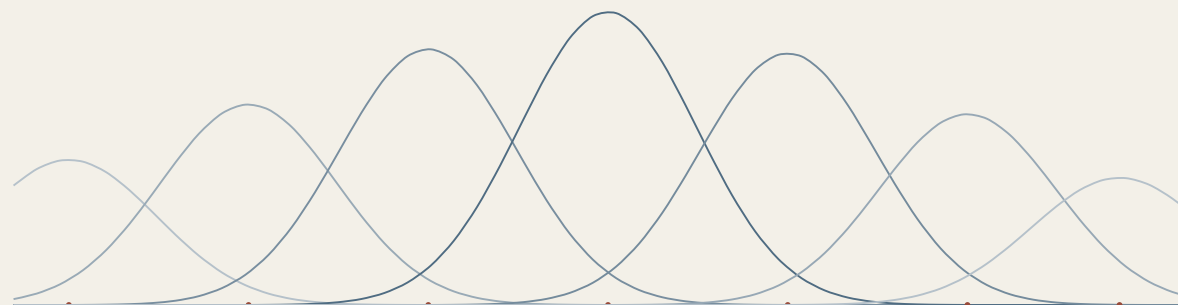
Kernels: The Geometry of Learning

Taha Bouhsine

Kernels

The Geometry of Learning

Foundations, algorithms, and the frontiers of machine intelligence



$$k(x, z) = \langle \Phi(x), \Phi(z) \rangle_{\mathcal{H}}$$

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